Dynamic Platoon Formation of Multi-Type Autonomous Vehicles for Sustainable Urban Mobility

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Abstract

We study an algorithmic abstraction for future cooperative autonomous mobility in which larger active vehicles can tow smaller passive vehicles along shared road segments. The core problem is spatio-temporal: given fixed vehicle trajectories, entry times, speeds, and active-vehicle towing capacities, determine which passive vehicles should be matched to which active vehicles, and over which route segments, to maximize total covered distance. Unlike standard platooning or one-to-one assignment, the problem allows multi-segment service, where a passive vehicle may be towed by different active vehicles over disjoint portions of its route, while active-vehicle capacity must be respected at every position along the corridor. We formulate this dynamic platoon formation problem and propose two approximation methods: a greedy maximum-weight matching baseline and an Iterative Linear Assignment method that expands active vehicles into virtual capacity slots and repeatedly solves linear sum assignment subproblems. Experiments on synthetic corridor scenarios with up to 400 passive and 80 active vehicles show that both methods achieve substantial towed-distance coverage (25–52% depending on capacity), with the assignment-based method consistently matching or exceeding greedy solution quality while reducing empirical runtime by 2.8–11.5×. We position the model as a corridor-level spatial matching primitive; mechanical coupling, safety, and full road-network routing are left as future extensions.

CCS Concepts

• **TODO: replace with CCS-generated concept → e.g. Theory of computation**.

Keywords

autonomous vehicles, vehicle platooning, spatial-temporal matching, linear assignment, sustainable mobility

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1 Introduction

Single-occupancy travel remains a major barrier to sustainable urban mobility. U.S. National Household Travel Survey data show that drive-alone commuting dominates trip demand [**?**], while the transportation sector contributes about 29% of U.S. greenhouse gas emissions [**?**]. This inefficiency is pronounced when many vehicles traverse overlapping routes but still consume propulsion energy independently. With rapid advances in autonomous driving [**?**], future transportation can realize physically coordinated vehicle platoons that go beyond independent driving.

In a near-future setting where vehicles are fully autonomous, new forms of physical coordination between vehicles become feasible: a large autonomous vehicle could mechanically couple with smaller vehicles on a shared road segment, towing them and offsetting a substantial share of their independent propulsion energy [**? ?**]. Realizing such coordinated interaction at scale requires *systematic decisions* about which vehicles should couple, when, and for how long along their routes. We therefore treat mechanical coupling, safety, and regulatory design as system assumptions outside the scope of this short paper, and focus on the spatio-temporal matching layer that would be required if such coupling were available.

To address this challenge we introduce a new problem that we call *dynamic platoon formation*. We consider two classes of autonomous vehicles: larger *Active Vehicles (AVs)* capable of physically towing smaller vehicles, and smaller *Passive Vehicles (PVs)* that can be attached to AVs during shared road segments. Unlike drafting-based platooning, which yields modest aerodynamic savings while vehicles remain physically separated, our model assumes mechanical coupling: while a PV is attached to an AV we model the PV as not self-propelled over that segment, with the segment's towing load attributed to the AV. Given a set of AVs and PVs with known, fixed routes (AVs do not deviate from their pre-planned paths), entry times, and speeds, the goal is to **maximize the total PV distance covered by towing**. We use towed distance as a proxy for PV-side propulsion-energy reduction and reserve physics-level load modeling for future work.

Figure 1 illustrates this on a shared corridor with two AVs and three PVs. Two features are central to our formulation: *point-wise capacity*—between x=20 and x=60, A_1 simultaneously tows $T1$ and $T2$, reaching its capacity of 2—and *multi-segment matching*, where

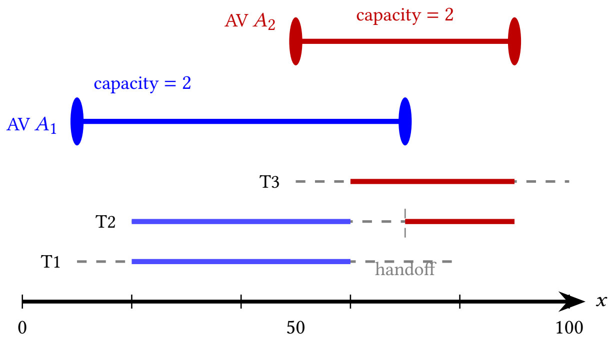

Figure 1: Conceptual overview of towing-based cooperation on a 1D corridor (positions 0–100 m). Dashed gray segments denote self-driving portions (PV uses own propulsion); solid colored segments denote towed travel (PV propulsion offset by AV). AVs A_1 and A_2 each have towing capacity 2. $T2$ is towed by A_1 over $[20, 60]$, self-drives over $[60, 70]$, and is handed off to A_2 over $[70, 90]$ – an instance of *multi-segment matching*.

$T2$ is served by A_1 and A_2 on disjoint portions of its route with a self-driving gap in between.

Prior work spans three threads. (i) Platooning research focuses on aerodynamic coordination among physically separated vehicles via cooperative adaptive cruise control [? ? ? ?]. (ii) Ridesharing and dial-a-ride methods match passengers or trips to vehicles under spatial and temporal constraints, but the decision variable is passenger placement rather than vehicle-to-vehicle coupling along route segments [? ? ? ?]. (iii) Combinatorial assignment provides algorithmic tools for one-to-one matching, including the Hungarian method and its extensions [? ?]. None of these lines captures the combination of features required for dynamic towing coordination: sustained physical coupling, multi-segment coverage of a single PV by different AVs, and *point-wise* AV capacity constraints—the requirement that the number of PVs an AV tows simultaneously is enforced at every position along its route and can change as PVs attach and detach.

We formulate the problem as: given AVs and PVs with fixed routes, entry times and speeds, decide which PVs attach to which AVs over which road segments to maximize the total towed PV distance, subject to spatial overlap requirements, point-wise capacity, and temporal synchronization at coupling points. This multi-segment matching problem generalizes the *Generalized Assignment Problem* (GAP), which is NP-hard [?]; exact solutions are intractable at realistic fleet sizes. We propose two approximations. The first, a *Greedy Maximum-Weight Matching*, repeatedly commits to the highest-saving feasible AV–PV segment until no feasible pair remains—fast but myopic. The second, an *Iterative Linear Assignment* (ILA) algorithm, takes a globally coordinated view: in each round it expands every AV into virtual capacity slots, builds a bipartite graph of feasible AV-slot–PV pairings, and solves the resulting *Linear Sum Assignment Problem* (LSAP) optimally via the Jonker–Volgenant method [?]. PV states are then updated to reflect newly covered segments and the round repeats until no further assignment is possible. For tractability we evaluate both algorithms on a one-dimensional corridor abstraction; we discuss extensions to 2D road networks in Section 5.

Contributions. (1) We formalize *dynamic platoon formation* as a spatio-temporal matching problem with multi-segment service and point-wise AV capacity, instantiated on a 1D corridor. **(2)** We develop and compare two approximation algorithms—a greedy maximum-weight baseline and an iterative LSAP-based method with virtual-slot capacity encoding. **(3)** On synthetic corridor scenarios with up to 80 AVs and 400 PVs, ILA matches or exceeds greedy solution quality (25–52% towed-distance coverage depending on capacity) while running 2.8–11.5× faster.

2 Problem Formulation

2.1 Notation and System Model

2.2 Shared Path and Energy Saving

2.3 Multi-Segment Matching and Point-wise Capacity

2.4 Optimization Problem

2.5 Assumptions

3 Proposed Algorithms

3.1 Greedy Maximum-Weight Matching

3.2 Iterative Linear Assignment (ILA)

4 Experimental Evaluation

4.1 Setup

4.2 Results

5 Conclusion and Future Work

References